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Concrete Experience

CAMOUFLAGE LEVEL:



EXPERT

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United States Army Cyber Center of Excellence

Establishing an Electromagnetic Warfare Hide Site

112-17ABOLC

OVERALL CLASSIFICATION:

UNCLASSIFIED



Learning Step Activities

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- LSA 1 - Identify Key Requirements and Considerations for an EW Observation Post (OP)
- LSA 2 - Identify Manning and Equipment Considerations for an EW OP
- LSA 3 - Examine Camouflage, Cover, and Concealment Considerations for an EW OP



Administrative Information



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Safety Requirements NONE

Risk Assessment Level LOW

Environmental Considerations None

Evaluation None



References

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- FM 3-12. *Cyberspace and Electromagnetic Warfare Operations*. 17 Sep 2025.
- FM 3-90. *Tactics*. 10 May 2023.
- ATP 3-12.3, C1. *Electromagnetic Warfare Techniques*. 18 Sep 2024.
- ATP 3-12.4. *Electromagnetic Warfare Platoon*. 09 Jan 2023.
- ATP 3-21.8, C1. *Infantry Rifle Platoon and Squad*. 23 Jun 2025.
- ATP 3-37.34/MCTP 3-34C. *Survivability Operations*. 16 Apr 2018.
- *TC 3-12.1.2.2. *The United States Army Cyber and Electromagnetic Warfare Training Strategy (CEWTS)*. 01 Nov 2024.



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Identify Key Requirements and Considerations for an EW Observation Post (OP)



What is an Observation Post?

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- A position from which observations are made, or fires are directed and adjusted.





Relevance to EW

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Question: Why do EW soldiers need to learn how to establish an observation post?

Answer: Allows EW PLTs to conduct successful surveillance of the EMS without being compromised.

Nesting within PLT METL

- Conduct Electromagnetic Support (ES)
- Electromagnetic Reconnaissance
- Establishing an observation post is a task for all reconnaissance tasks.

Bottom Line: Every EW team should be expected to know how establish an observation post.



Selection of the Observation Post



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- Information requirement >>> develop plan to obtain the information
- Observation plan is part of COA sketch.
- Done by asking two basic questions:
 - What is the best location to obtain the information required?
 - What is the best way to obtain the information without being compromised?
- The EW PLT leadership determines how many OPs are needed
 - Positioning of each OP
 - Allow long-range observation of the selected area



Characteristics of an Observation Post



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- Covered and concealed routes to and from each position.
- Unobstructed observation of the assigned area, route, or zone.
- Fields of observation of adjacent positions overlap, to ensure full coverage.
- Cover and concealment reduce the OPs vulnerability on the battlefield





Characteristics of an Observation Post



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- Positions should not be sited in such locations.
- Do not skyline the observers.
- Avoid hilltops.
- Locate positions farther down the slope of the hill or on the side





Occupation of the Observation Post



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- Moving to the OP, an appropriate technique is used based on mission analysis
- To occupy:
 - Conduct a leader's recon
 - Occupy it by force
- Securing the site is the priority before occupation.
 - Weapons systems are emplaced
 - 360-degree security





Occupation of the Observation Post



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- Security Complete >>> Conduct an EMS Survey to ensure that it has EMS observation of the target/area.
- Conduct an EMS Survey >>> EW Observation Post established





Mobility Considerations



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- Dismounted
 - Advantages: stealth, less terrain restrictions
 - Disadvantages: limited flexibility and limited firepower/equipment
- Mounted
 - Advantages: flexibility, additional equipment can be carried
 - Disadvantage: less stealth, more hindered by terrain

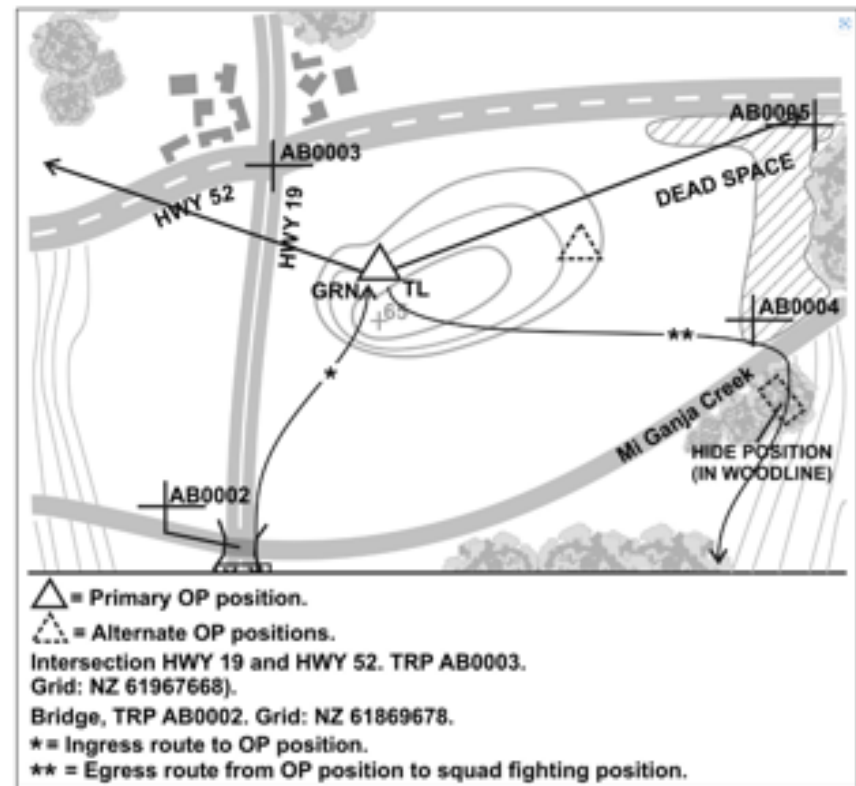




Drawing A Sketch

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- The sketch will include:
 - Key and significant terrain
 - Location of the observation post
 - Alternate observation post
 - Routes to the OPs and fighting positions
 - Sectors of observation.
 - Direct and indirect fire control measures.





Practical Exercise (Drawing A Sketch)



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Task: Prepare Sketch of the surrounding environment

Conditions: Given a writing utensils and a paper and an identified sector in an outdoor environment

Standards: Within 15 minutes draw a sketch that depicts

- A rough sketch of key and significant terrain
- Your location
- Routes
- Estimation how far you can see
- Dead space



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Identify Manning and Equipment Considerations for an EW OP



Manning/Equipping the Observation Post



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- Minimum 2 Soldiers
 - 12 hours or more require team/squad
- Rotation plans
- Equipment for observation posts:
 - Map of the area
 - Compass/GPS
 - Communications equipment
 - Observation devices





Manning/Equipping the Observation Post



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- Report formats
- Weapon systems
- Protective obstacles
- Early warning devices
- Camouflage
- Deception equipment





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Examine Camouflage, Cover, and Concealment Considerations for an EW OP



Camouflage, Cover and Concealment



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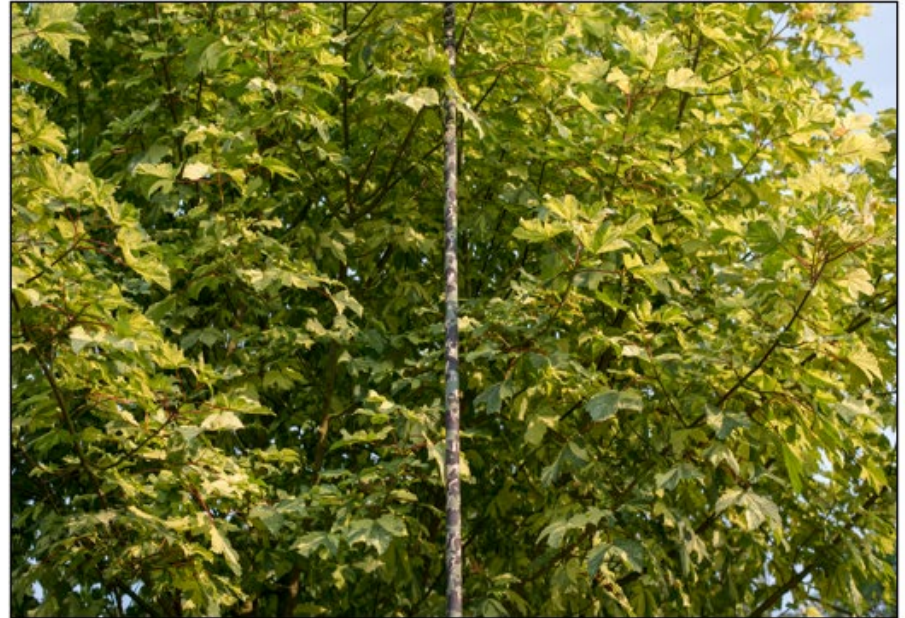
- Camouflage and cover/concealment is priority for the success of an observation post.
- Unique equipment of the EW PLT makes them identifiable both visually and in the EMS.
- EW PLTs have very limited defensive capability





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Camouflage, Cover and Concealment





Visual Camouflage and Cover/Concealment



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- Visual Camouflage: Man made equipment look distinctly different than its natural surroundings.
- So what: Visual observation is the quickest way for an observation post to become decisively engaged through accurate fire.
- Which means: You need to utilize camouflage, cover, and concealment to reduce or eliminate the enemy's ability to see your observation post.





Visual Camouflage and Cover/Concealment



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- Infiltration/Exfiltration Routes: Routes used to enter and exit an observation post.
- So what: Moving to and from your OP is a time of great risk.
- Which means: You need routes that are cover and concealed and generally away from natural lines of drift.



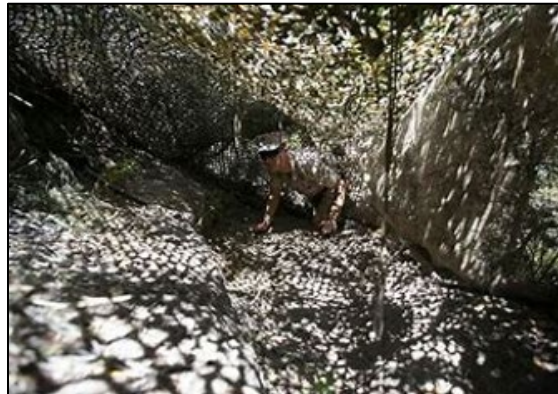


Visual Camouflage and Cover/Concealment



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- Hide Site: A concealed area in which you place equipment and other support equipment that is not immediately needed on the observation post.
- So what: The more equipment you have at the observation post, the bigger a target the observation post becomes.
- Which means: Place unnecessary equipment or sleep/eat areas away from the observation post in a concealed area to reduce risk of compromise.

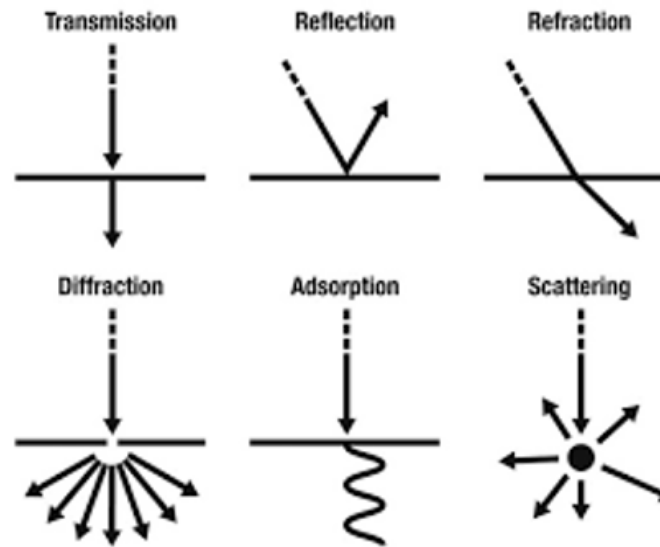


EMS Camouflage and Cover/Concealment



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- EM Waves interact with the world in six ways
 - Transmission
 - Reflection
 - Refraction
 - Diffraction
 - Absorption
 - Scattering
- Manipulated radio signatures can become directional
- Interactions with other metallic objects can change an antennas radiation pattern

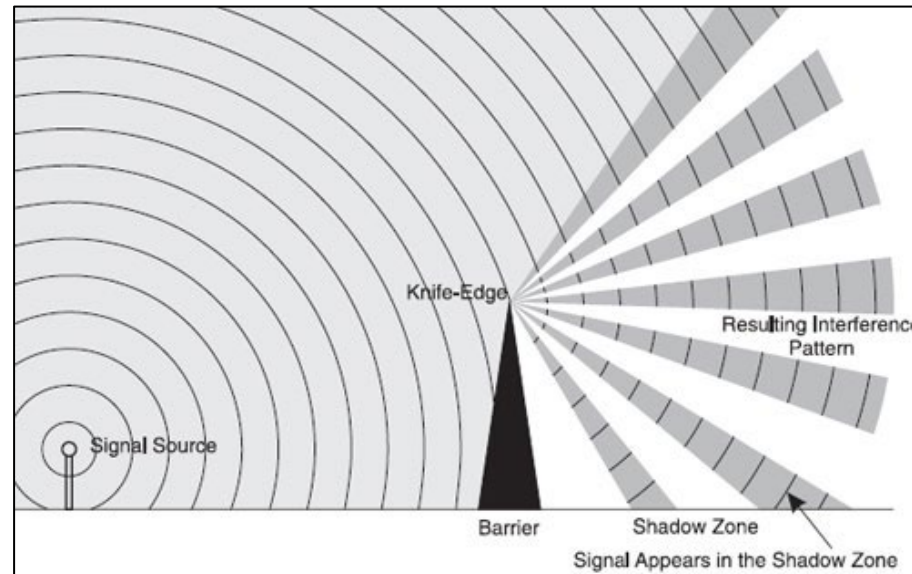




Knives Edge Diffraction

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- Knives Edge Diffraction: Occurs when a signal bends around an obstacle and is received on the other side as an interference pattern
- So what: This means you do not have to have visual line of site to observe a signal in the EMS.
- Which means: You can be physically covered and still observe your target

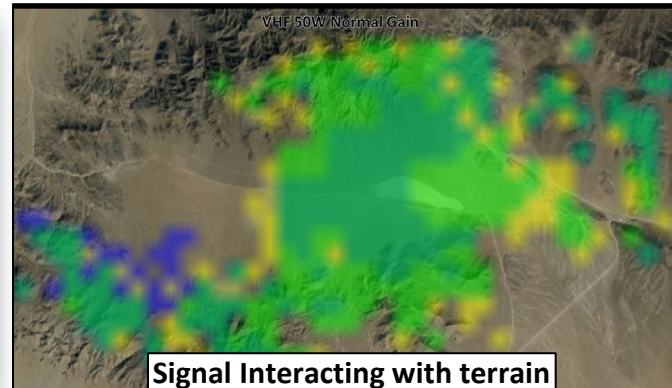




Terrain Masking

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- Terrain Masking: Occurs when a signal is unable to be seen past one or more natural obstacles due to the combined effects of absorption and reflection
- So what: Terrain can stop the enemy from sensing signals coming from your observation post.
- Which means: Place your communications equipment behind terrain so only your force can see the signals



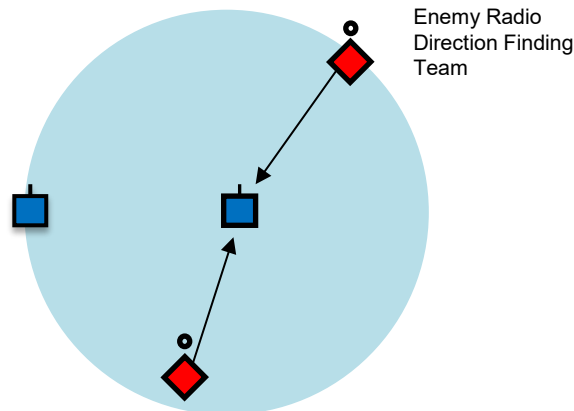


Directional Antennas

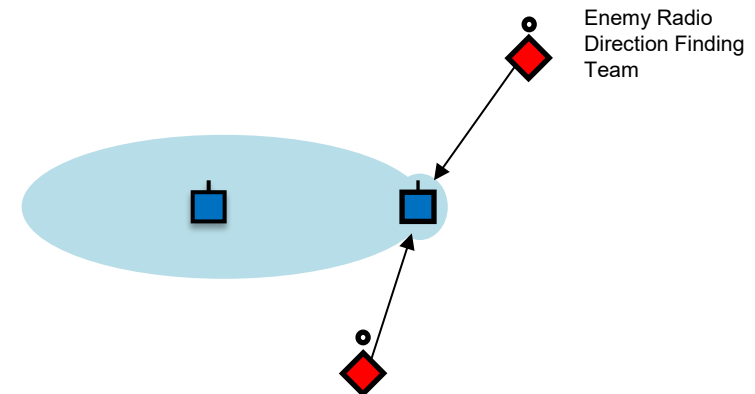
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- Directional Antennas: Antennas that have that have a bias to receive and transmit signals in a particular direction or pattern
- So what: Directional antennas can decrease interception of your signals.
- Which means: You can talk to friendly forces with a large decrease of being detected by the enemy.

Omnidirectional Antennas



Directional Antennas





Remoting Antennas

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- Remoting Antennas: Antennas can be separated from the radio by wireless or wired means.
- So what: All reception and transmitting of a signal happens at the antenna.
- Which means: You increase survivability without sacrificing capability of your observation post by keeping your soldiers and equipment behind cover and remoting your antennas.

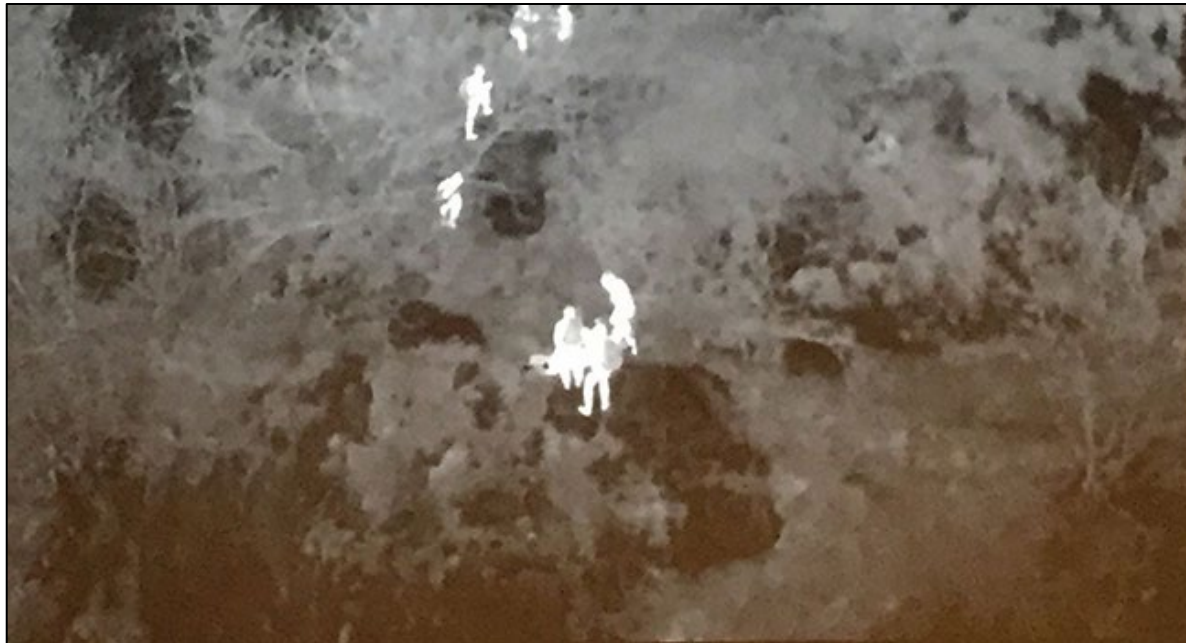




Thermal Signatures

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- Thermal Signatures: Equipment and people give off heat.
- So what: Heat can be seen in the Infrared portion of the EMS
- Which means: You need to be aware of thermal management.





Site Establishment Group Discussion



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- Task 1: Take 10 Mins to read through the Table 3 Task 5: Conduct EW Site Establishment from the Cyber and Electromagnetic Warfare Training Strategy (CEWTS).
- Task 2: Take 10 Mins to discuss in groups of 3-5, what important tasks you identified and what planning/preparation inputs you would need in order to do those tasks i.e. (products, equipment PCCs and PCIs...)



Conduct EW Site Establishment



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- **Condition:** Members of the EW PLT with their assigned EW equipment and a selected and approved mounted or dismounted EW sites.
- **Standard:** Upon arrival at your approved EW site. Re-verify the feasibility of the EW site. Relocate as necessary. You will have so much time to completely recover your EW systems, tear down the EW site, and be ready for movement.
- **Evaluator Execution:** Present the Soldier with the task, condition, and standard and instruct them to execute. The EW PLT will conduct movement to their primary EW site chosen in the first task of this table. The EW PLT will then establish their EW site and prepare their EW system for operations.



Conduct EW Site Establishment



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1. Receive the mission and conduct TLPs
 - a. Conduct mission planning (integrate into supporting units' scheme of maneuver)
 - b. Identify methods of reducing EMI to key civilian infrastructure if applicable
 - c. Conduct PCCs and PCIs
 - d. Prepare EW equipment for operation
 - e. Configure EW equipment IAW mission tasking
2. Confirm PACE plan (CST and supported unit)



Conduct EW Site Establishment



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3. Enforce EP measures
 - a. Provide recommendations to supporting unit (EMCON, equipment vulnerabilities, system redundancy)
4. Maneuver to and establish EW site IAW table 3 task 9
5. Conduct EW IAW the AGM
 - a. Verify adjacent SIGINT/ES asset observation of EA effects
 - b. Confirm target frequency parameters and target acquisition
 - c. Reappraise and apply jamming equation
 - d. Attack the target (review jamming control, program jam power output, and jam target through predetermined techniques)
 - e. Displace to alternate locations IAW displacement criteria



Conduct EW Site Establishment



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6. Determine Re-Attack recommendation (observe change of enemy use of the EMS)
7. Assess EP measures
8. Recover equipment and sanitize EW site
9. Provide EWMSNSUM report to CST upon completion of mission



Summary

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- Identified Key Requirements and Considerations for an EW Observation Post (OP)
- Identified Manning and Equipment Considerations for an EW OP
- Examined Camouflage, Cover, and Concealment Considerations for an EW OP



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Questions